Today

- More on properties of solutions of time-independent Schrödinger equation

1.15.4 Spatial part of stationary wavefunction can be chosen to be real

Given $\psi_E(\vec{r})$ is a solution to the time-independent Schrodinger equation with energy E,

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r}) + V(\vec{r})\psi_E(\vec{r}) = E\psi_E(\vec{r}), \tag{1.15.6}$$

and complex conjugating it, we have

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r})^* + V(\vec{r})\psi_E(\vec{r})^* = E\psi_E(\vec{r})^*. \tag{1.15.7}$$

In other words, $\psi_E(\vec{r})^*$ is also a solution with the same energy. Therefore, we can trade $\psi_E(\vec{r})$ and $\psi_E(\vec{r})^*$ for two other combinations,

$$\psi_E(\vec{r}) + \psi_E(\vec{r})^* \qquad \text{and} \qquad \frac{\psi_E(\vec{r}) - \psi_E(\vec{r})^*}{i}, \tag{1.15.8}$$

which are both real. Thus, the spatial part of a stationary wavefunction can be chosen to be real.

1.15.5 Parity symmetry: Even and odd wavefunctions

Suppose the potential is symmetric under reflection about the origin, $V(-\vec{r}) = V(\vec{r})$. In this case, given

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r}) + V(\vec{r})\psi_E(\vec{r}) = E\psi_E(\vec{r}), \tag{1.15.9}$$

and replacing $\vec{r}$ by $-\vec{r}$ and setting $V(-\vec{r}) = V(\vec{r})$, we have a

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(-\vec{r}) + V(\vec{r})\psi_E(-\vec{r}) = E\psi_E(-\vec{r}). \tag{1.15.10}$$

Therefore, $\psi_E(-\vec{r})$ is a solution with the same energy. We can trade $\psi_E(\vec{r})$ and $\psi_E(-\vec{r})$ for two other combinations,

$$\psi_E(\vec{r}) + \psi_E(-\vec{r}) \qquad \text{and} \qquad \psi_E(\vec{r}) - \psi_E(-\vec{r}). \tag{1.15.11}$$

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Now $\psi_E(\vec{r}) + \psi_E(-\vec{r})$ is even under parity transformation, since $\psi_E(\vec{r}) + \psi_E(-\vec{r}) \to \psi_E(-\vec{r}) + \psi_E(\vec{r}) = \psi_E(\vec{r}) + \psi_E(-\vec{r})$ under $\vec{r} \to -\vec{r}$, while $\psi_E(\vec{r}) - \psi_E(-\vec{r})$ is odd with $\psi_E(\vec{r}) + \psi_E(-\vec{r}) \to \psi_E(-\vec{r}) - \psi_E(\vec{r}) = -(\psi_E(\vec{r}) + \psi_E(-\vec{r}))$. Thus, when the potential is symmetric under reflection through the origin, we can choose the stationary states to be either even or odd under reflection.

1.15.6 Orthogonality/orthonormality

Suppose $\psi_n(\vec{r}, t)$ is a solution with energy E_n and $\psi_m(\vec{r}, t)$ is a solution with energy E_m; i.e.,

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_n(\vec{r}) + V(\vec{r})\psi_n(\vec{r}) = E_n\psi_n(\vec{r}) \quad \text{and} \quad -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_m(\vec{r})^* + V(\vec{r})\psi_m(\vec{r})^* = E_m\psi_m(\vec{r})^*, \tag{1.15.12}$$

where we have taken the complex conjugate for the second. Multiplying both sides of the first by $\psi_m(\vec{r})^*$ and the second by $\psi_n(\vec{r})$,

$$\begin{aligned} -\frac{\hbar^2}{2m}\psi_m(\vec{r})^*\vec{\nabla}^2\psi_n(\vec{r}) + V(\vec{r})\psi_m(\vec{r})^*\psi_n(\vec{r}) &= E_n\psi_m(\vec{r})^*\psi_n(\vec{r}) \\ -\frac{\hbar^2}{2m}\psi_n(\vec{r})\vec{\nabla}^2\psi_m(\vec{r})^* + V(\vec{r})\psi_n(\vec{r})\psi_m(\vec{r})^* &= E_m\psi_n(\vec{r})\psi_m(\vec{r})^* \end{aligned} \tag{1.15.13}$$

Subtracting the two,

$$-\frac{\hbar^2}{2m}\left(\psi_m(\vec{r})^*\vec{\nabla}^2\psi_n(\vec{r}) - \psi_n(\vec{r})\vec{\nabla}^2\psi_m(\vec{r})^*\right) = (E_n - E_m)\psi_m(\vec{r})^*\psi_n(\vec{r}) \tag{1.15.14}$$

Recalling our computation and definition of the probability current, the left hand side equals $-i\hbar\vec{\nabla}\cdot\vec{J}$, where $\vec{J} = -\frac{i\hbar}{2m}\left(\psi_m(\vec{r})^*\vec{\nabla}\psi_n(\vec{r}) - \psi_n(\vec{r})\vec{\nabla}\psi_m(\vec{r})^*\right)$. Therefore,

$$-i\hbar\vec{\nabla}\cdot\vec{J} = (E_n - E_m)\psi_m(\vec{r})^*\psi_n(\vec{r}). \tag{1.15.15}$$

Integrating both sides, and using the divergence theorem, $\int \vec{\nabla}\cdot\vec{J}d^3\vec{r} = \oint \vec{J}\cdot d\vec{a}$,

$$-i\hbar\oint \vec{J}\cdot d\vec{a} = (E_n - E_m)\int \psi_m(\vec{r})^*\psi_n(\vec{r})d^3\vec{r}. \tag{1.15.16}$$

Now, taking the boundary of the area integral to infinity, the current vanishes, since a normalizable wavefunction and the corresponding current must vanish at infinity, and the left hand

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side vanishes. We then have

$$(E_n - E_m) \int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = 0. \tag{1.15.17}$$

We have two cases.

1. Case 1: $n = m$ and $E_n = E_m$. In this case, $E_n - E_m = 0$ and for a normalized wavefunction $\psi_n(\vec{r})$, $\int \psi_n(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = 1$.
2. Case 2: $n \neq m$ and $E_n \neq E_m$. In this case, $E_n - E_m \neq 0$ and $\int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = 0$. We say $\psi_n(\vec{r})$ and $\psi_m(\vec{r})$ are orthogonal.

Combining the two cases, we have the orthonormality relation

$$\int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = \delta_{mn}, \tag{1.15.18}$$

where, for the Kronecker delta, now

$$\delta_{mn} = \begin{cases} 1, & m = n \\ 0, & m \neq n. \end{cases}$$

1.15.7 Linearity

The Schrödinger equation is linear. Given stationary states $\psi_n(\vec{r}, t) = \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$, $n = 1, 2, ...$, a linear combination of these

$$\psi(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}, \tag{1.15.19}$$

where c_n are complex constants, is a solution to the time-dependent Schrödinger equation $i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = \hat{H}\psi(\vec{r}, t)$. To check that, the left hand side is $i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = \sum_n c_n \psi_n(\vec{r}) i\hbar \frac{\partial}{\partial t} e^{-\frac{i}{\hbar} E_n t} = \sum_n c_n E_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$, and the right hand side is $\hat{H}\psi(\vec{r}, t) = \sum_n c_n \hat{H}\psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t} = \sum_n c_n E_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$, and they are equal term by term.

1.15.8 Time evolution

Question 1.15.2: Given stationary states $\psi_1(\vec{r}, t)$ and $\psi_2(\vec{r}, t)$ with different energies, is $c_1\psi_1(\vec{r}, t) + c\psi_2(\vec{r}, t)$ stationary? (A). Yes, (B). No.

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Answer: The answer is (B). No. That is because each of $\psi_1(\vec{r},t)$ and $\psi_2(\vec{r},t)$ oscillate in time with different frequencies and $|c_1\psi_1(\vec{r},t)+c\psi_2(\vec{r},t)|^2$ depends on time. Setting $c_1=c_2=\frac{1}{\sqrt{2}}$ and choosing $\psi_1(\vec{r})$ and $\psi_2(\vec{r})$ to be real for simplicity, $|\frac{1}{\sqrt{2}}(\psi_1(\vec{r},t)+\psi_2(\vec{r},t))|^2=|\frac{1}{\sqrt{2}}e^{-\frac{i}{\hbar}E_1}(\psi_1(\vec{r})+\psi_2(\vec{r})e^{-\frac{i}{\hbar}(E_2-E_1)}|^2=\frac{1}{2}((\psi_1(\vec{r}))^2+(\psi_2(\vec{r}))^2+2\psi_1(\vec{r})\psi_2(\vec{r})\cos(\frac{E_2-E_1}{\hbar})$. Thus, the state oscillate with angular frequency $\omega=\frac{E_2-E_1}{\hbar}$.

Because $\psi(\vec{r},t)=\sum_n c_n\psi_n(\vec{r},t)=\sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}$ with constant coefficients c_n is a solution to the time-dependent Schrödinger equation, and $\psi(\vec{r},0)=\sum_n c_n\psi_n(\vec{r},0)=\sum_n c_n\psi_n(\vec{r})$ at time $t=0$, we note that given

$$\psi(\vec{r},0)=\sum_n c_n\psi_n(\vec{r},0)=\sum_n c_n\psi_n(\vec{r}) \tag{1.15.20}$$

at time $t=0$, the time evolution, or the wavefunction at time t, is simply

$$\psi(\vec{r},t)=\sum_n c_n\psi_n(\vec{r},t)=\sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}. \tag{1.15.21}$$

In other words, once we have expanded a given initial wavefunction in terms of a linear combination of the stationary wavefunctions $\psi_n(\vec{r})$, the time evolution follows simply by putting a factor of $e^{-\frac{i}{\hbar}E_nt}$ to each term containing $\psi_n(\vec{r})$. We have already checked above following Eq. (1.15.21) that this solves the time-independent Schrödinger equation.

1.15.9 Normalization

Given $\psi(\vec{r},t)=\sum_n c_n\psi_n(\vec{r},t)=\sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}$ with complex coefficients c_n, normalizing it leads to

$$\begin{aligned}1 &= \int|\psi(\vec{r},t)|^2d^3\vec{r}=\int\left(\sum_m c_m^*\psi_m^*(\vec{r})e^{\frac{i}{\hbar}E_mt}\right)\left(\sum_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}\right)d^3\vec{r}\\ &= \sum_n\sum_m c_m^*c_ne^{\frac{i}{\hbar}(E_m-E_n)t}\int\psi_m^*(\vec{r})\psi_n(\vec{r})d^3\vec{r}=\sum_n\sum_m c_m^*c_ne^{\frac{i}{\hbar}(E_m-E_n)t}\delta_{mn}=\sum_n|c_n|^2.\end{aligned} \tag{1.15.22}$$

Therefore, the constant coefficients are constrained by

$$\sum_n|c_n|^2=1. \tag{1.15.23}$$

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1.15.10 Completeness

Given an arbitrary $\psi(\vec{r},t)$ that is a solution to the Schrödinger equation with certain boundary conditions, we can express it as a linear superposition of the stationary solutions $\psi_n(\vec{r},t) = \psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t}$ as

$$\psi(\vec{r},t) = \sum_n c_n \psi_n(\vec{r},t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar}E_n t}. \tag{1.15.24}$$

with complex coefficients c_n. In other words, the stationary orthonormal solutions form a complete set.

To find the condition for the above statement to hold, first let us find the coefficients c_n. Multiplying both sides by $\psi_m(\vec{r},t)^*$ and integrating over $d^3\vec{r}$,

$$\int \psi_m(\vec{r},t)^* \psi(\vec{r},t) d^3\vec{r} = \sum_n c_n \int \psi_m(\vec{r},t)^* \psi_n(\vec{r},t) d^3\vec{r} = \sum_n c_n \delta_{mn} = c_m. \tag{1.15.25}$$

Therefore,

$$c_n = \int \psi_n(\vec{r},t)^* \psi(\vec{r},t) d^3\vec{r}. \tag{1.15.26}$$

Inserting these coefficients back into the expansion,

$$\begin{aligned} \psi(\vec{r},t) &= \sum_n \int \psi_n(\vec{r}',t)^* \psi(\vec{r}',t) d^3\vec{r}' \psi_n(\vec{r},t) \\ &= \int d^3\vec{r}' \psi(\vec{r}',t) \sum_n \psi_n(\vec{r}',t)^* \psi_n(\vec{r},t). \end{aligned} \tag{1.15.27}$$

The stationary states form a complete set if

$$\sum_n \psi_n(\vec{r}',t)^* \psi_n(\vec{r},t) = \delta^3(\vec{r}' - \vec{r}), \tag{1.15.28}$$

where $\delta^3(\vec{r}' - \vec{r})$ is the Dirac-delta function in three dimensions defined by

$$\int d^3\vec{r}' \psi(\vec{r}',t) \delta^3(\vec{r}' - \vec{r}) = \psi(\vec{r},t). \tag{1.15.29}$$

In other words, if we multiply and sum the stationary states as $\sum_n \psi_n(\vec{r}',t)^* \psi_n(\vec{r},t)$ and we get $\delta^3(\vec{r}' - \vec{r})$, we know we have a complete set. Conversely, if we miss a state in the sum, we do not get $\delta^3(\vec{r}' - \vec{r})$. In cartesian coordinates, $\delta^3(\vec{r}' - \vec{r}) = \delta(x - x')\delta(y - y')\delta(z - z')$.

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1.16 Delta function

Below, we summarize some of the properties of the delta function, focusing on one dimension. Given a function $f(x)$, the delta function $\delta(x-x')$ is defined such that

$$f(x') = \int f(x)\delta(x-x')dx \tag{1.16.1}$$

In other words, $\delta(x-x')$ in the integral projects out the value of $f(x)$ at the zero of the argument $x-x'$ of $\delta(x-x')$. Therefore, we can view $\delta(x-x')$ as defined by

$$\delta(x-x') = 0 \text{ for } x \neq x', \qquad f(x') = \int_a^b f(x)\delta(x-x')dx \text{ for } a < x' \text{ and } b > x'. \tag{1.16.2}$$

Because $\delta(x-x') = 0$ for $x \neq x'$, we can rewrite $f(x') = \int f(x)\delta(x-x')dx = f(x')\int \delta(x-x')dx$. Therefore,

$$\int \delta(x-x')dx = 1. \tag{1.16.3}$$

However, the delta function is not actually a function but a distribution. For instance, we can not interpret $\int \delta(x-x')dx = 1$ as the area under $\delta(x-x')$ being one.

Here we list some representations of the delta function,

$$\delta_\alpha(x) = \frac{1}{\alpha\sqrt{\pi}} e^{-\frac{x^2}{\alpha^2}} \tag{1.16.4}$$

$$\delta_\alpha(x) = \frac{1}{\pi x}\sin\left(\frac{x}{\alpha}\right) \tag{1.16.5}$$

$$\delta_\alpha(x) = \frac{\alpha}{\pi x^2}\sin^2\left(\frac{x}{\alpha}\right) \tag{1.16.6}$$

With any of these representations, the delta function $\delta_\alpha(x-a)$ is defined through

$$f(x') = \lim_{\alpha\to 0}\int f(x)\delta_\alpha(x-x')dx. \tag{1.16.7}$$

However, $\lim_{\alpha\to 0}\int f(x)\delta_\alpha(x-x')dx \neq \int \lim_{\alpha\to 0} f(x)\delta_\alpha(x-x')dx$. Indeed the integral vanishes if the limit is taken within the integral.

Exercise 1.16.1: Show the following.

1. $\delta(ax) = \frac{1}{|a|}\delta(x)$

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2. $\delta(x^2 - a^2) = \frac{1}{2|a|}(\delta(x + a) + \delta(x - a))$
3. $\delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{|f'(x)|}$, where x_i are the roots of $f(x) = 0$.
4. Write $\delta^3(\vec{r} - \vec{r}')$ in spherical coordinates.
5. Write $\delta^3(\vec{r} - \vec{r}')$ in cylindrical coordinates.

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